

Definition 2: If f is a function from A to B , we say that A is the domain of f and B is the codomain of f . If $f(a) = b$, we say that b is the image of a and a is a preimage of b . The range, or image, of f is the set of all images of elements of A . Also, if f is a function from A to B , we say that f maps A to B .

Remark: Note that the codomain of a function from A to B is the set of all possible values of such a function (that is, all elements of B), and the range is the set of all values of $f(a)$ for $a \in A$, and is always a subset of the codomain.

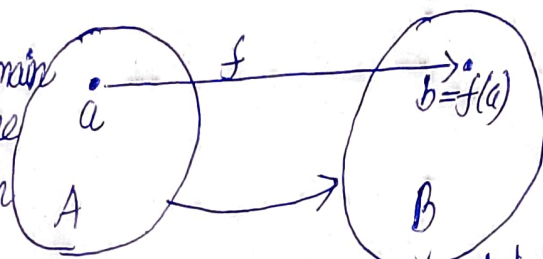


Figure 2 The function f maps A to B

When we define a function we specify its domain, its codomain, and the mapping of elements of the domain to elements in the codomain.

Two functions are equal when they have the same domain, have the same codomain, and map each element of their common domain to the same element in their common codomain.

Example 1: What are the domain, codomain, and range of the function that assigns grades to students described in the first paragraph of the introduction of this section?

Solution: Domain: $\{\text{Adam, Chou, Goodfriend, Rodriguez, Stevens}\}$.
Codomain: $\{A, B, C, D, F\}$

The range of G is the set $\{A, B, C, F\}$, because each grade except D is assigned to some student. $\square$

Example 2: Let R be the relation with ordered pairs $(\text{Abdul}, 22)$, $(\text{Brenda}, 24)$, $(\text{Carla}, 21)$, $(\text{Desire}, 22)$, $(\text{Eddie}, 24)$, and $(\text{Felicia}, 22)$. Here each pair consists of a graduate student and this student's age. Specify a function determined by this relation.

Solution: Domain: $\{\text{Abdul, Brenda, Carla, Desire, Eddie, Felicia}\}$
Co Domain: $\{x: x \in \mathbb{N}, x/18 \text{ and } x \leq 100\} \rightarrow$ Set of all ages in between 18 and 100.